AmazingMart EU

Optimizing Discount Strategy to Improve Profitability

A Data-Driven Analysis of Discount Policies, Margin Performance & Profit Recovery

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DASHBOARD: AMAZINGMART EU

CONTROLLED DISCOUNT ORDERS (≤20%) **6,621**

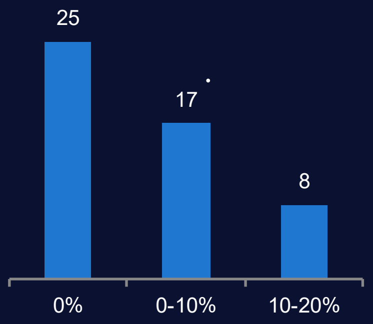
UNCONTROLLED DISCOUNT ORDERS (>20%) **1,426**

OVERVIEW

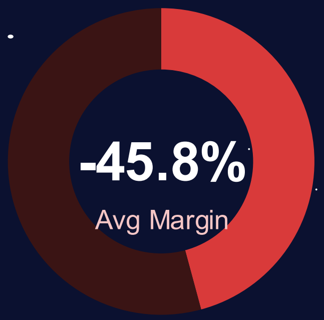
8%
Avg Discount

€2.03M
Total Sales

Margin % by discount band



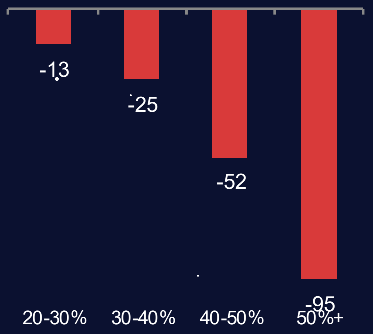
OVERVIEW



41%
Avg Discount

-€144K
Total Loss

Margin % by discount band



ANALYSIS METHODS APPLIED

Correlation	Regression	P&L Simulation	Country Breakdown	Elasticity Check
r = 0.004	p = 0.74	€169K	5 markets	No effect

OUR ANALYSIS & RECOMMENDATIONS WILL HELP AMAZINGMART EU RECOVER €169K IN ANNUAL PROFIT (+60%)

APPROACH FOR ANALYSIS

OVERALL APPROACH OF THE ANALYSIS

DATA EXTRACTION (SQL)	EDA (Tableau/Excel)	STATISTICAL TESTING (Python)	P&L SIMULATION	RECOMMENDATIONS

ABOUT AMAZINGMART EU

8,047	€2.35M	1,426	-€144K	€169K
Order lines analyzed (2011-2014)	Total revenue generated	Orders discounted over 20%	Loss from those orders	Recoverable annual profit

Source: MySQL extraction of order-level transaction data

1 Overview: OVERALL APPROACH

- Hypothesis Testing — correlation & regression to test if discount depth drives order volume.
- Policy Simulation — reconstructed list price and cost per order to simulate profit under a capped-discount policy.

Analysis considered order-level Discount, Quantity, Sales, Profit, Country, and Category fields

Statistical Methods

- Pearson correlation & linear regression
- P&L simulation via list-price reconstruction
- Country-level recovery cross-check

Dashboard built in Tableau, segregated into Controlled vs. Uncontrolled discount views

EDA Tools

- SQL (MySQL) & Excel
- Python (pandas, scipy)
- Tableau

INSIGHTS: Discount-volume independence · loss concentrated in 5 countries

RECOMMENDATION: 20% discount cap

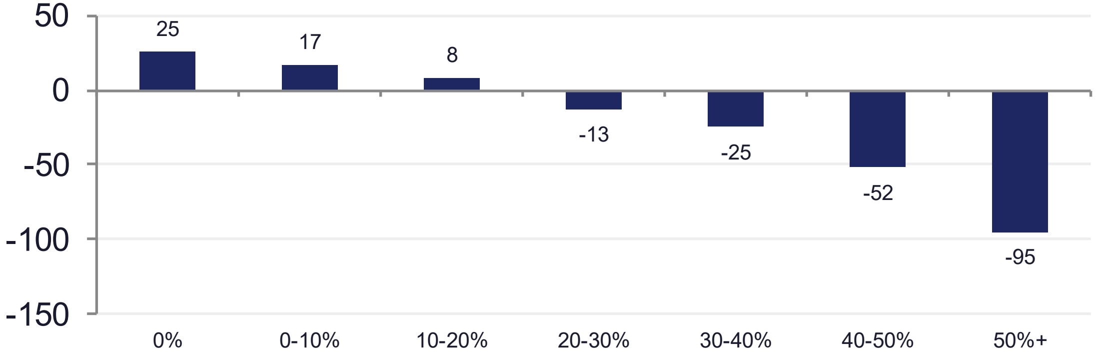
DATA PREPARATION & VALIDATION

CLEANING, JOINING, AND VALIDATING THE ANALYSIS

1 MAJOR STEPS INVOLVED

Joined line-item orders to order headers on Order ID

Margin % by discount band (order-level, n=8,047)



- Reconstructed List Price = Sales / (1 - Discount) to isolate the effect of discount from underlying price
- Derived Cost = Sales - Profit, held constant across simulation scenarios
- Cleaned encoding issues in Country/Region text fields prior to database import

2 VALIDATING THE ASSUMPTION

Tested whether volume held constant across discount depth

AVG. UNITS PER ORDER, BY DISCOUNT BAND

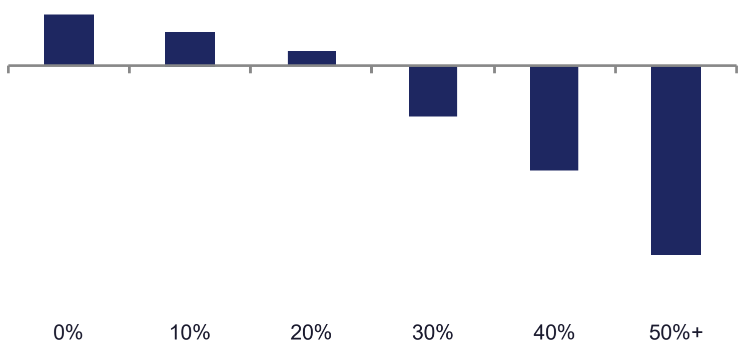
Discount Band	Avg Qty / Order
0%	3.78
0-10%	3.74
10-20%	3.72
20-30%	3.50
30-40%	3.70
40-50%	3.82
50%+	3.86

Order size stays flat (~3.5-3.9 units) regardless of discount depth — confirming discounting does not buy incremental volume (r=0.004, p=0.74).

EXPLORATORY DATA ANALYSIS

VISUALIZING THE DATA AND HIGHLIGHTING THE STRIKING INSIGHTS

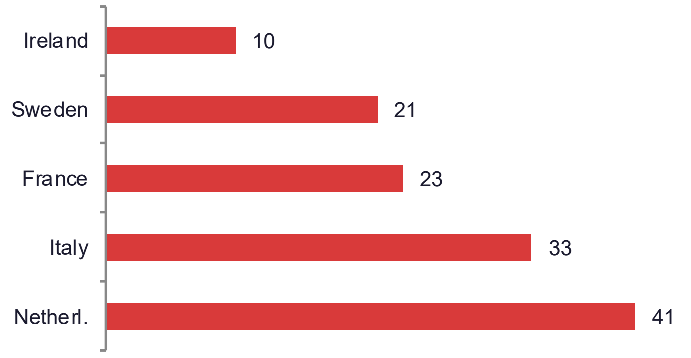
1 DISCOUNT IMPACT



MARGIN COLLAPSES PAST 20% DISCOUNT

- Margin turns negative beyond a 20% discount, reaching -94.6% past 50%.
- No corresponding rise in units purchased at any discount depth.

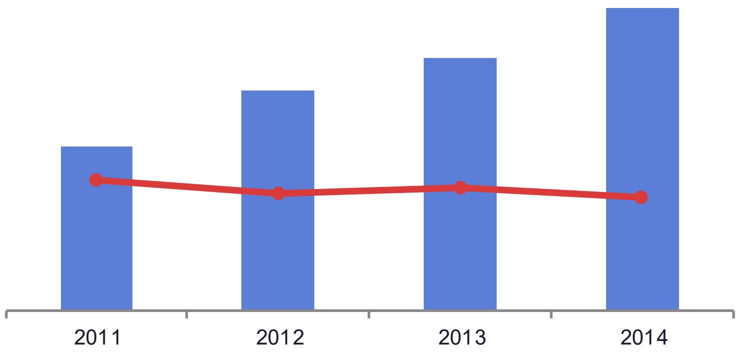
2 COUNTRY CONCENTRATION



LOSSES CONCENTRATED IN 5 COUNTRIES

- Netherlands, Sweden, Portugal, Ireland & Denmark average ~49% discount vs. 7% elsewhere.
- These 5 markets capture the majority of the recoverable €169K.

3 YEARLY TREND



GROWTH IS BEING BOUGHT WITH MARGIN

- Sales nearly doubled 2011-2014, while margin declined from 13.2% to 11.3%.
- Revenue growth alone masks the underlying profitability erosion.

A 20% discount cap is projected to recover €169K in annual profit (+60%), concentrated in 5 identified markets — with negligible volume risk.